

Data Flow Diagrams

Lecture 6

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Data Flow Diagrams (1)

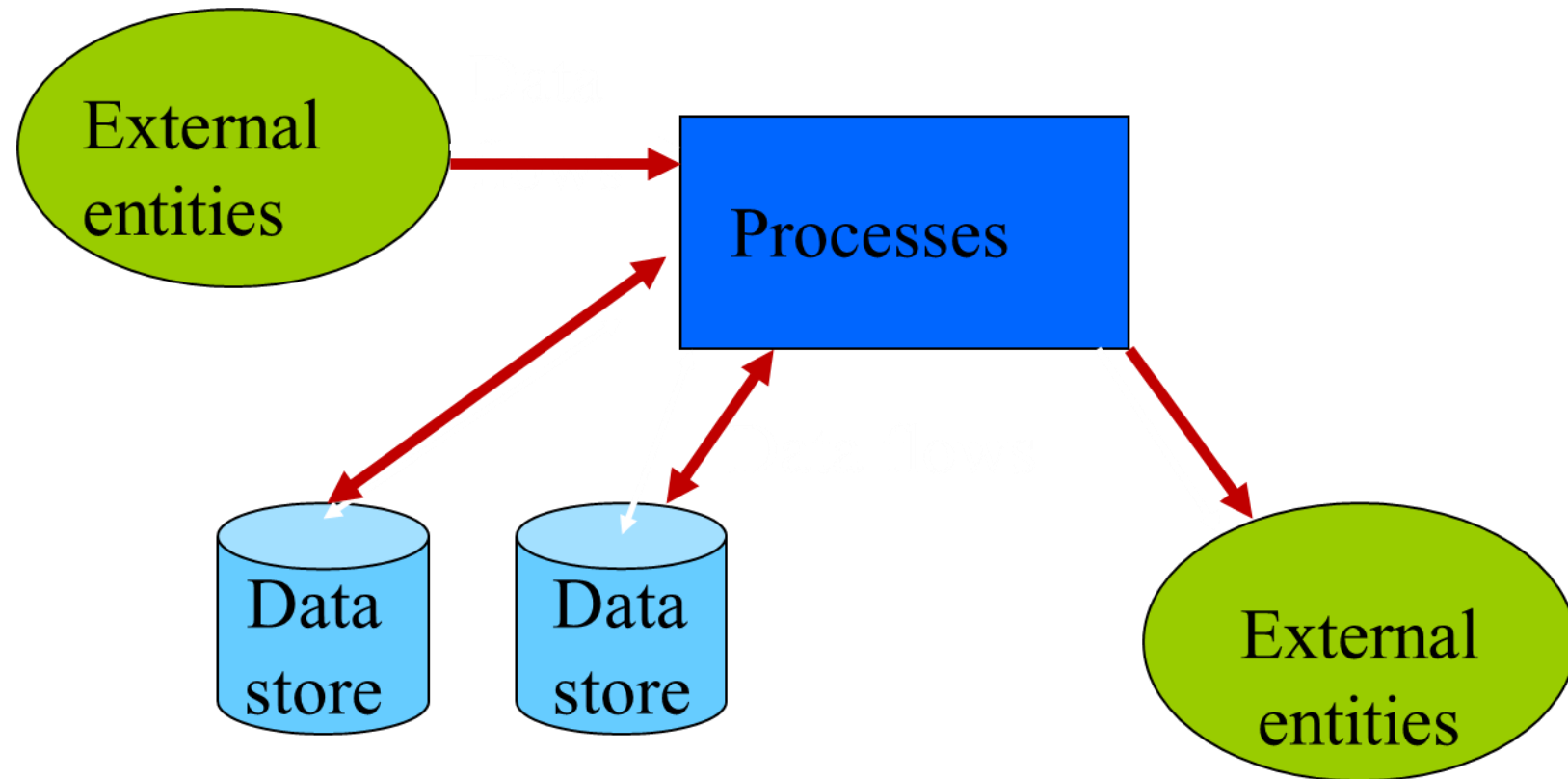
- Data Flow Diagrams (DFD) are using often to model the software requirements.
- It is a way of documenting the flow of data through a system through graphical means.
- Helps Analyst in the development with an understanding of the system
 - Convey the analyst's ideas to others.

Data Flow Diagrams (2)

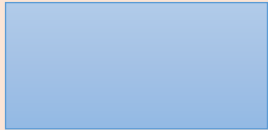
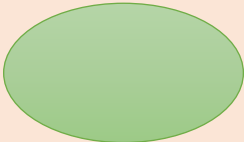
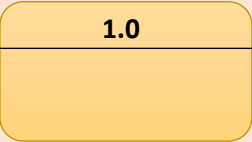
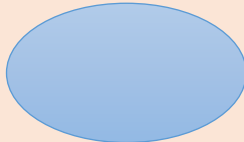
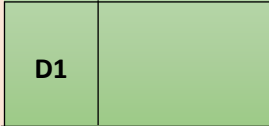
- A DFD is a diagrammatic representation of the information flows within a system showing
 - What information enters and leaves the system
 - What changes the information (processes)
 - Where information is stored
 - Who donates and receive information from the system
- Data Flow Diagrams are an important technique for systems analysis inside SSADM (Structured Systems Analysis and Design Method).

Data Flow Diagrams (3)

- Components

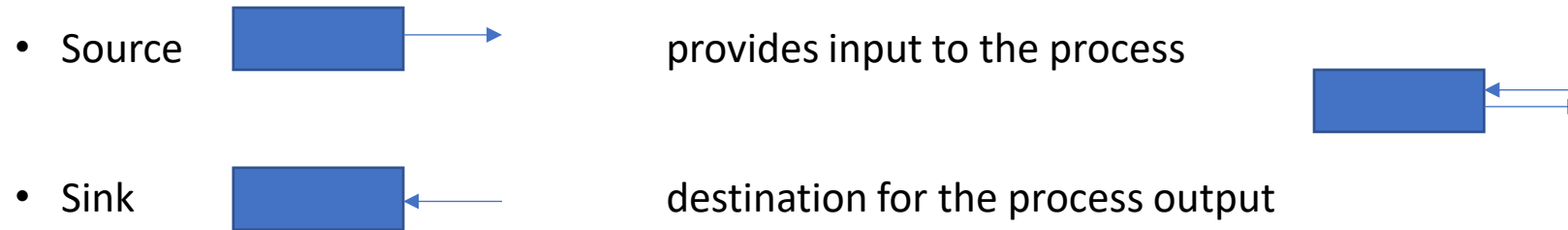


Data Flow Diagram Notations

	Yourdon and De Marco	Gane and Sarson	SSADM	Yourdon and Coad
Entity				
Process				
Data Store				
Data Flow				

External Entities

- Outside objects interact with the system.
 - Person
 - Organization
 - System
- External Entity is whoever provides information/data to the system or receives information/data from it.

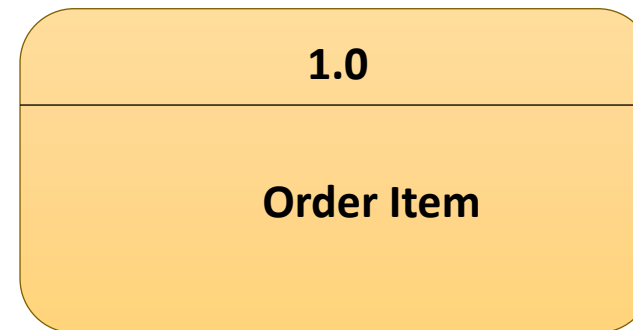


- Should be labeled as a noun
 - Ex: Customer
- Represented as a rectangle containing the name and an identifier.

Processes

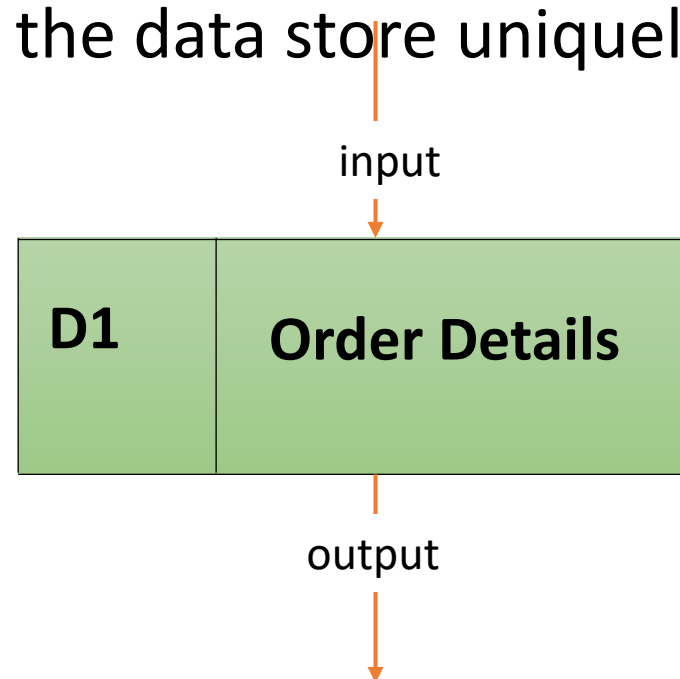
- A process is an activity which can be manual or automated which transform incoming data into outputs.
- Labeling
 - Reference Number
 - Process Name as Verb/ Noun

1.0 – First Process and Level 0
DFD



Data Stores

- Data stores is where the information held within the system.
- Each data store must have input and output data flows.
- Label must be a noun phrase.
- Unique identifier to identify the data store uniquely.
 - D – Electronic data source
 - M – Manual data source
 - T – Transient data source



Data Flow

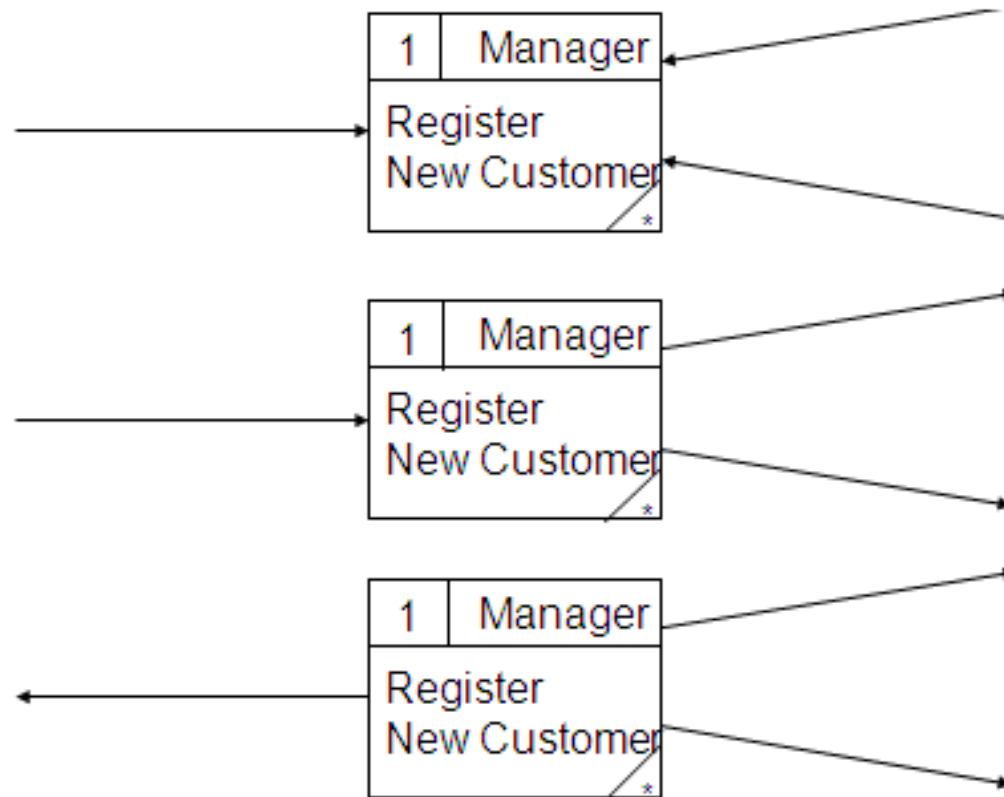
- A passage of information flowing between objects on the Data Flow Diagram represented an arrow to denote the direction of the information flow.
- Labeled the arrow with a name or details of the information represented by the data flow.
- Labeled as nouns.



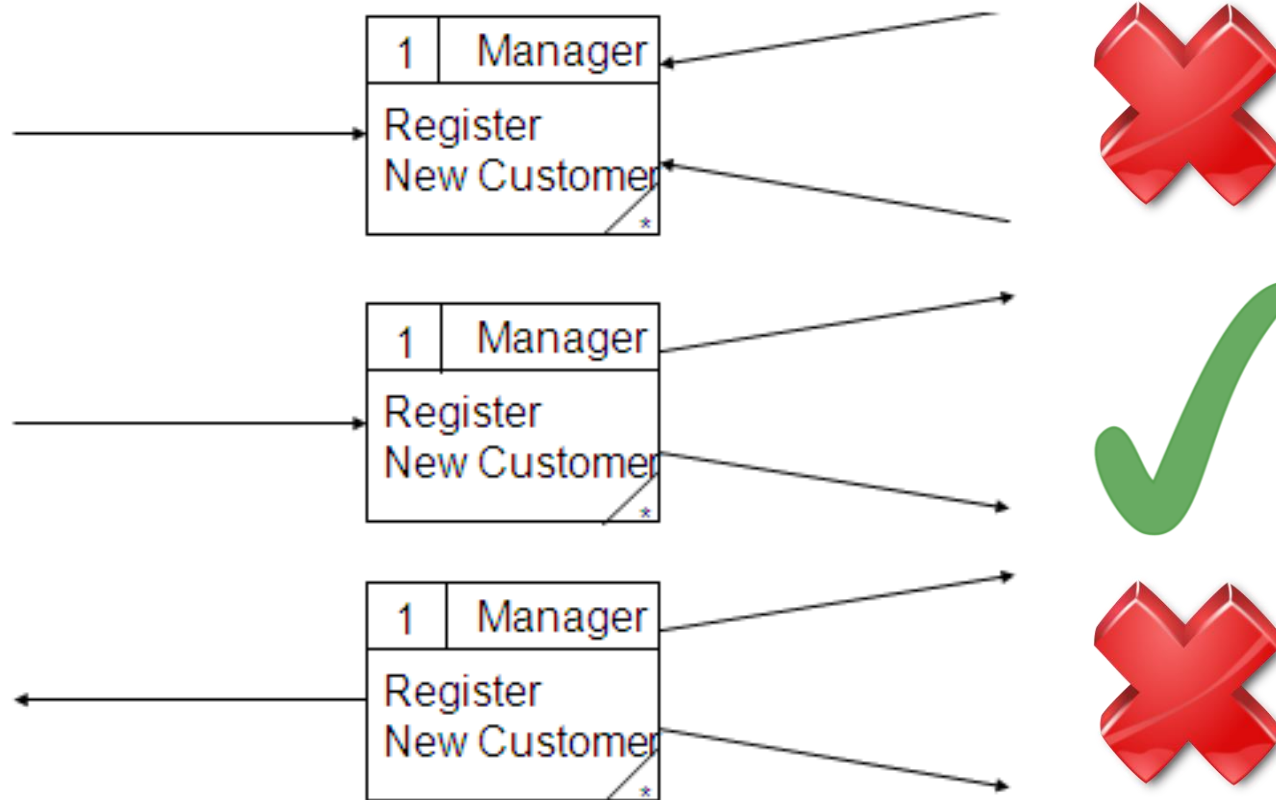
DFD- Rules and Best Practices (1)

- Every process must have at least one input data flow and one output data flow.
- Every external entity and data-store must have at least one output data flow, or input data flow.
- For every data flow either its source (originate) is a process or its destination is a process.

DFD- Rules and Best Practices (2)



DFD- Rules and Best Practices (3)

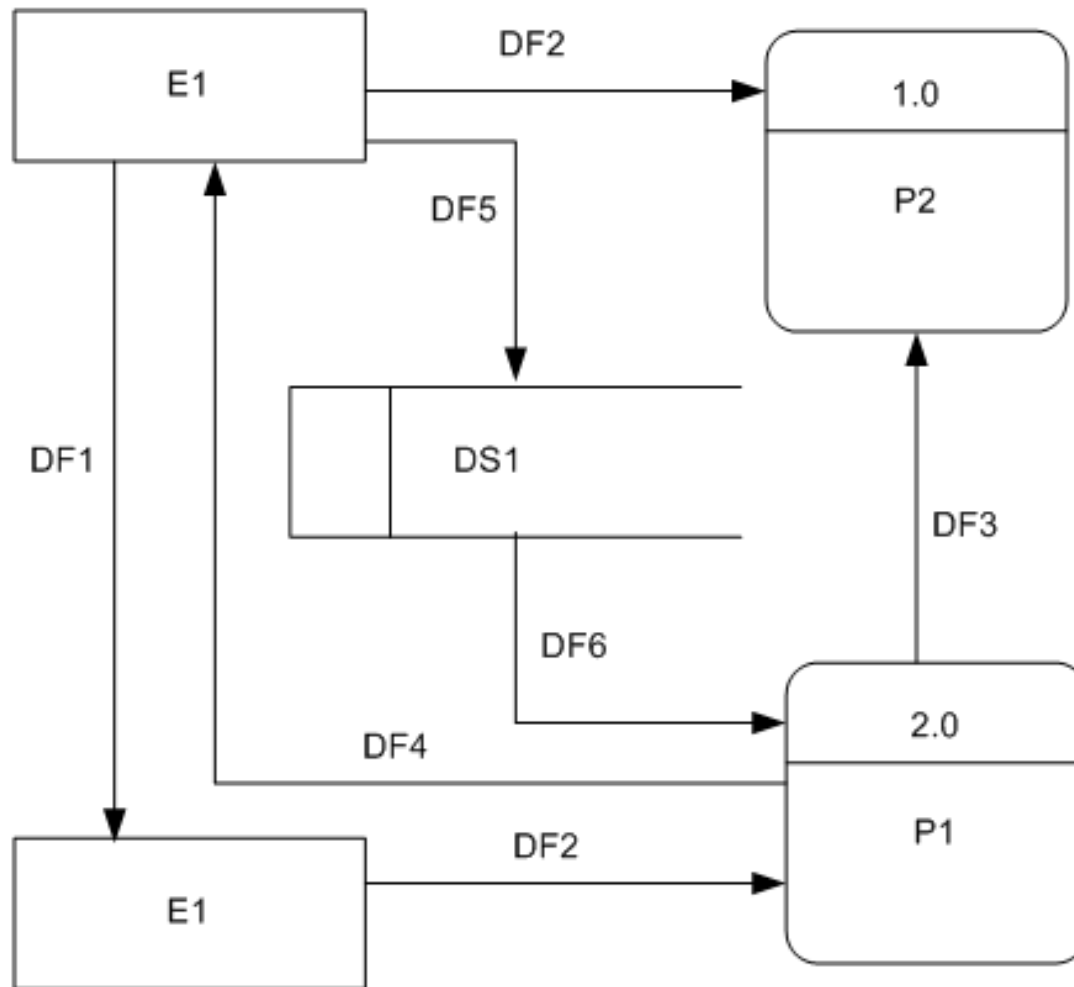


DFD- Rules and Best Practices (4)

Data flow can connect

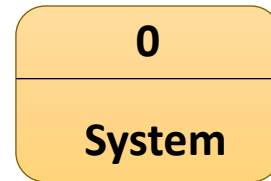
	YES	NO
A process to another process	✓	
A process to an external entity	✓	
A process to a data store	✓	
An external entity to another external entity		✓
An external entity to a data store		✓
A data store to another data store		✓

List the errors of this DFD

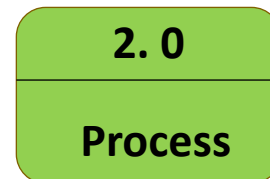
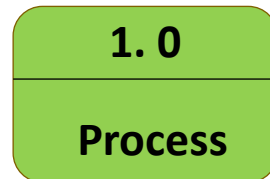


Levels of Data Flow Diagram

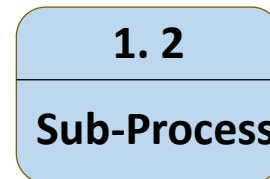
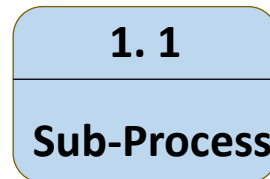
- Context Diagram



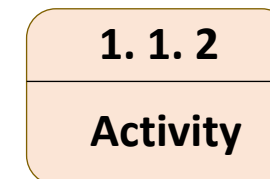
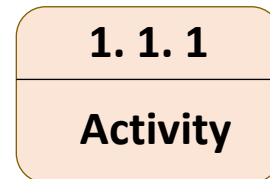
- Level 1



- Level 2

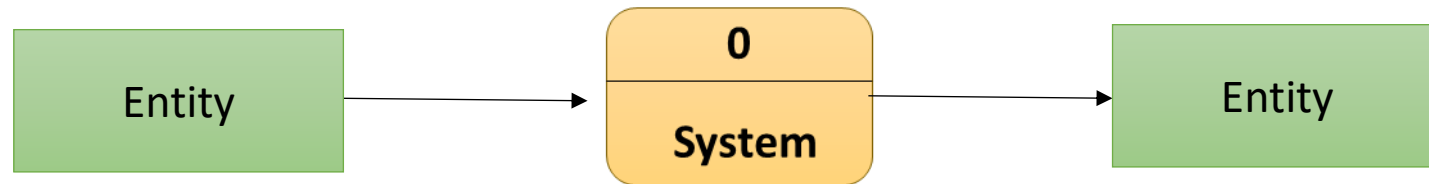


- Level 3

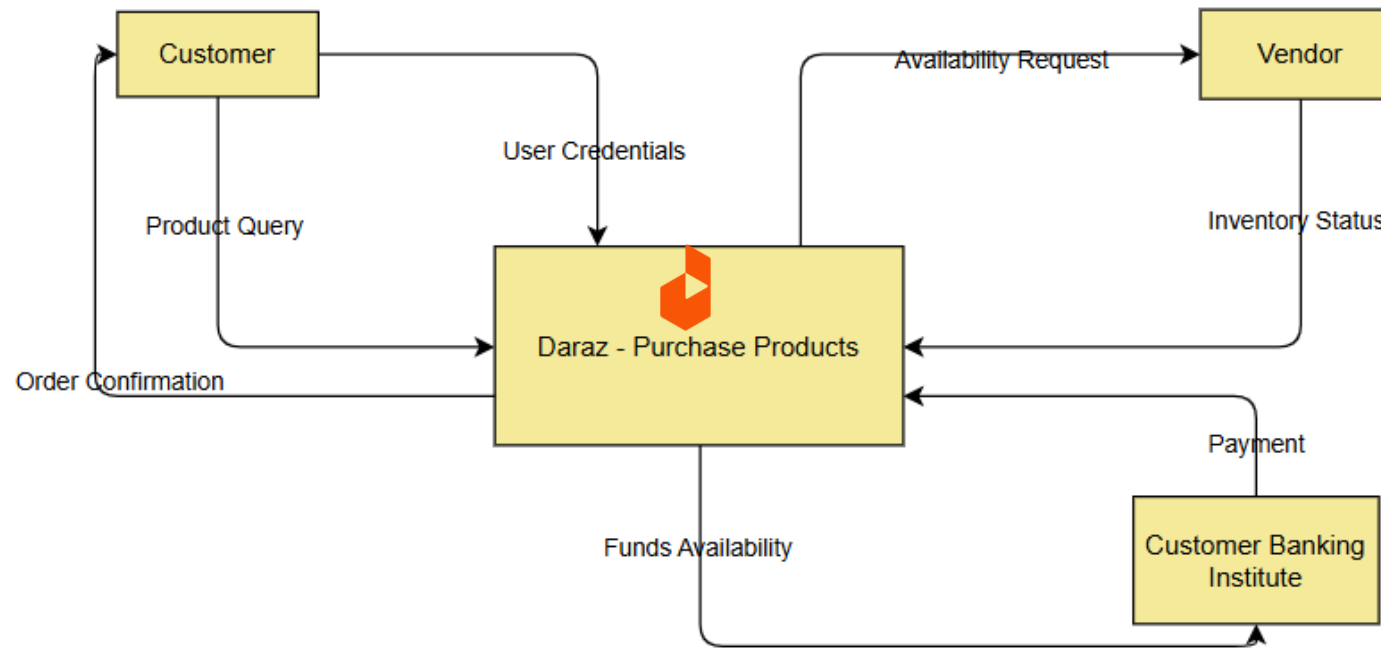


Context Diagram (1)

- Context diagram is an overview of the system that show the system boundaries, external entities that interact with the system, and the major information flows between the external entities and the system.



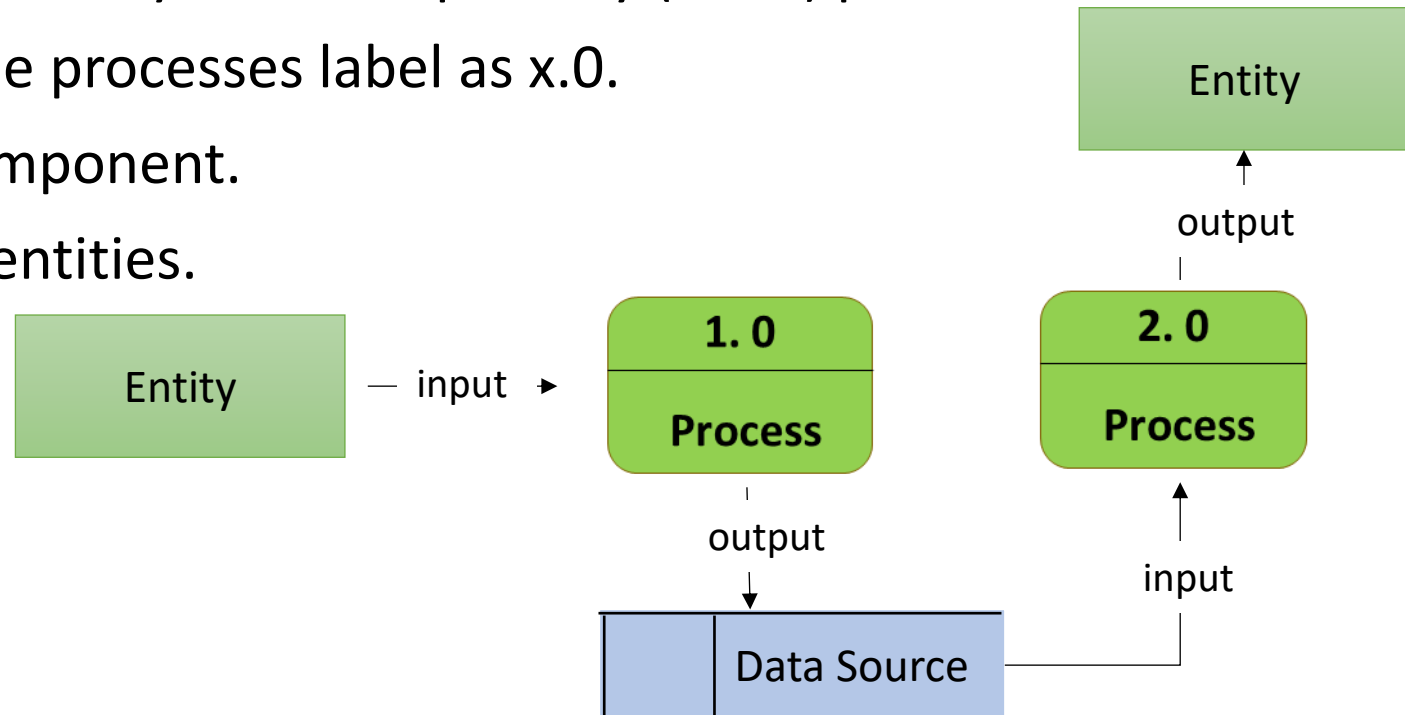
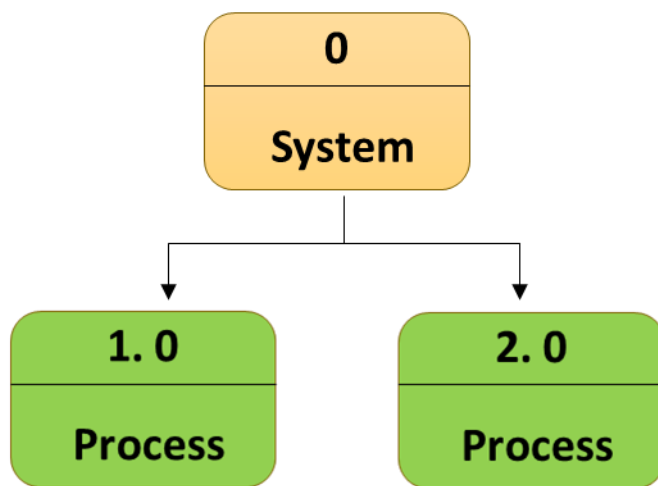
Context Diagram (2) - Example



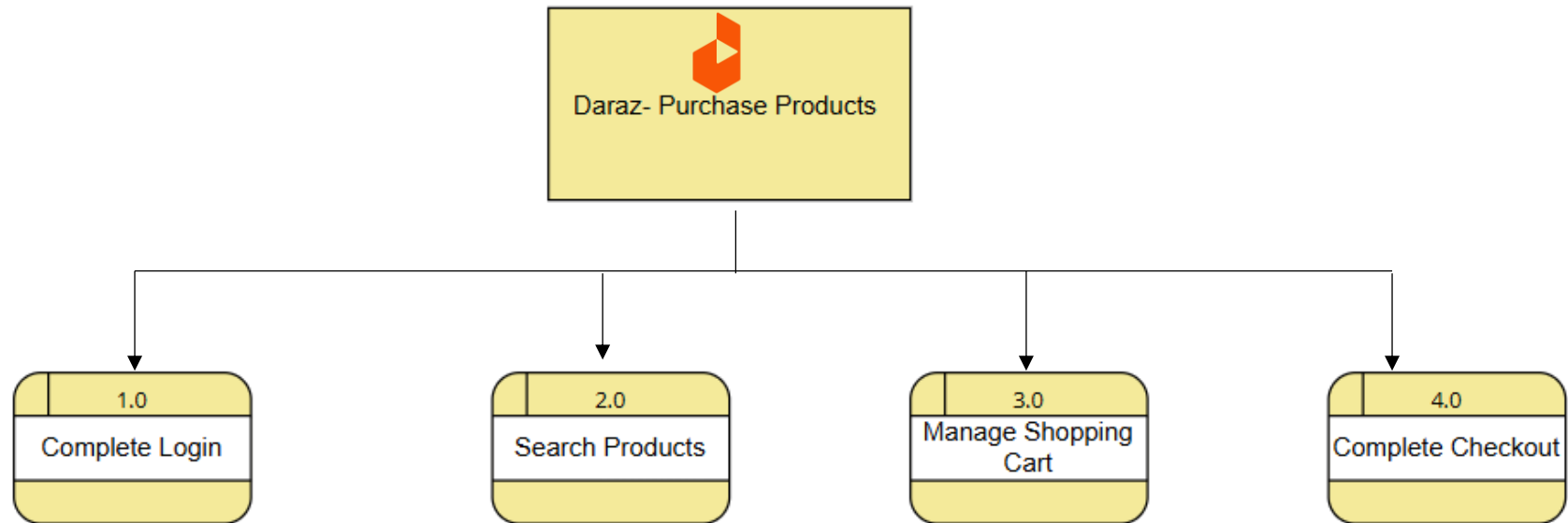
Note: This simulation only covers simple parts of the Daraz app, not the complete system. For simplicity, we have considered only a few of the main data flows.

Level 1

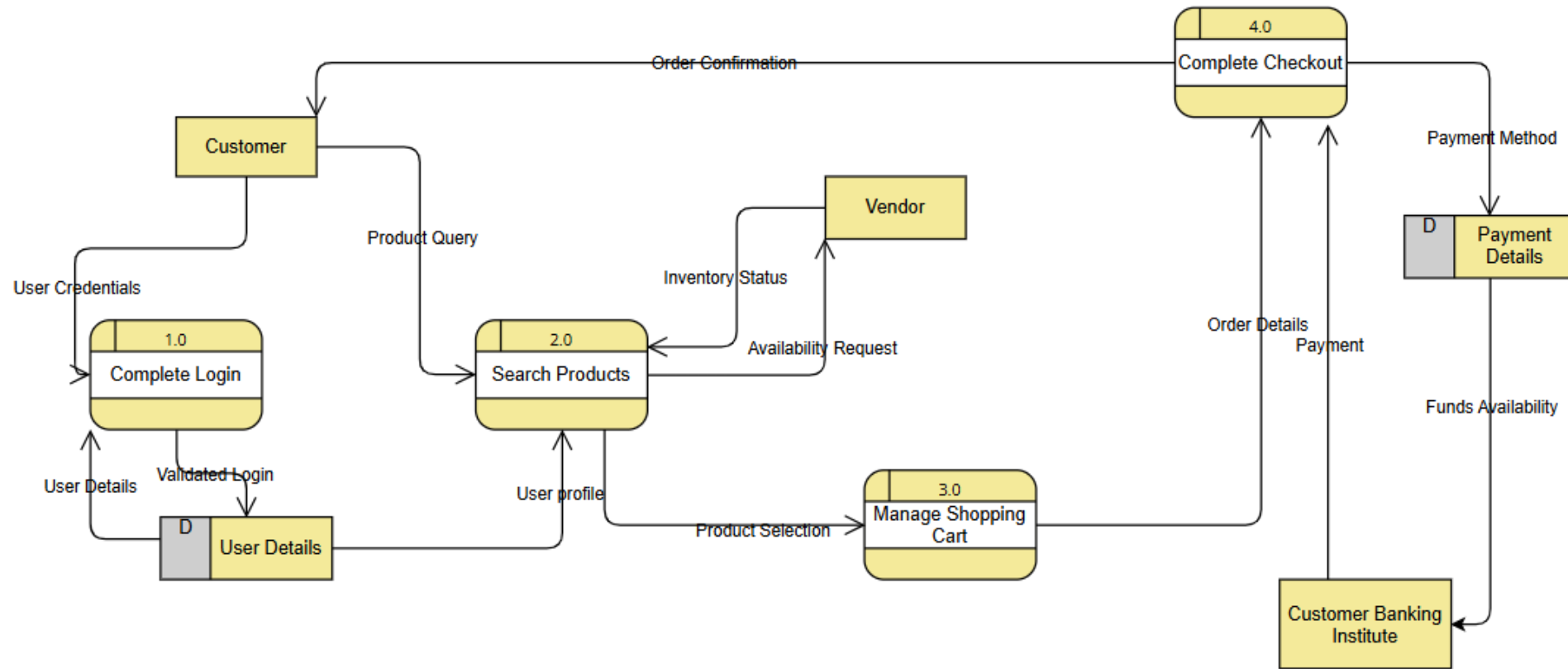
- This level decomposes entire system into primary (main) processes.
- Reference numbers of the processes label as x.0.
- Introduces data store component.
- Maintain same external entities.



Level 1 – Example (1)

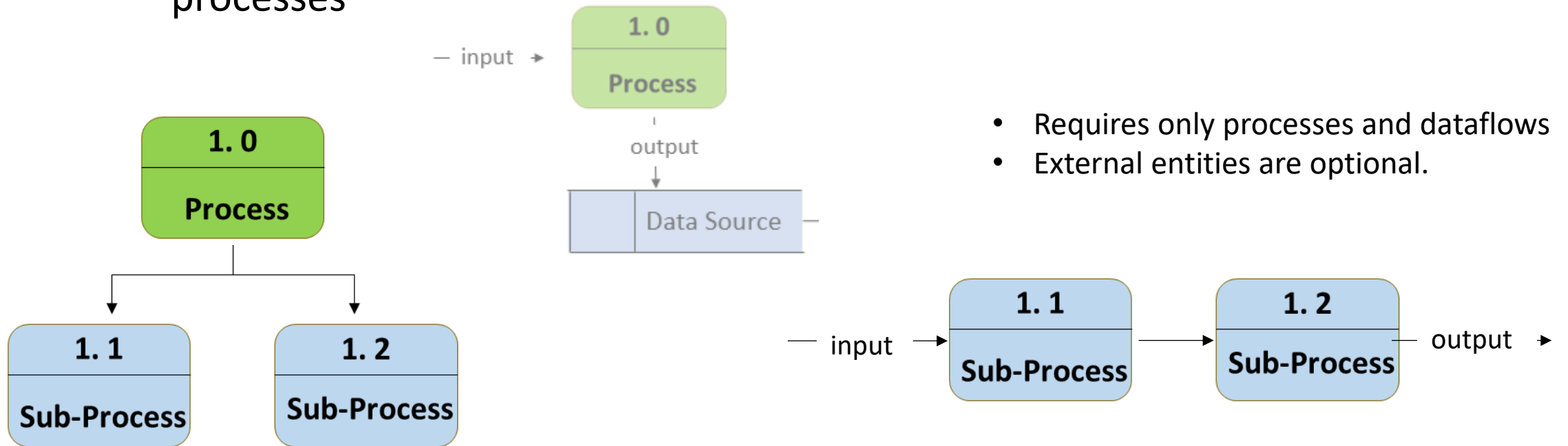


Level 1 – Example (2)

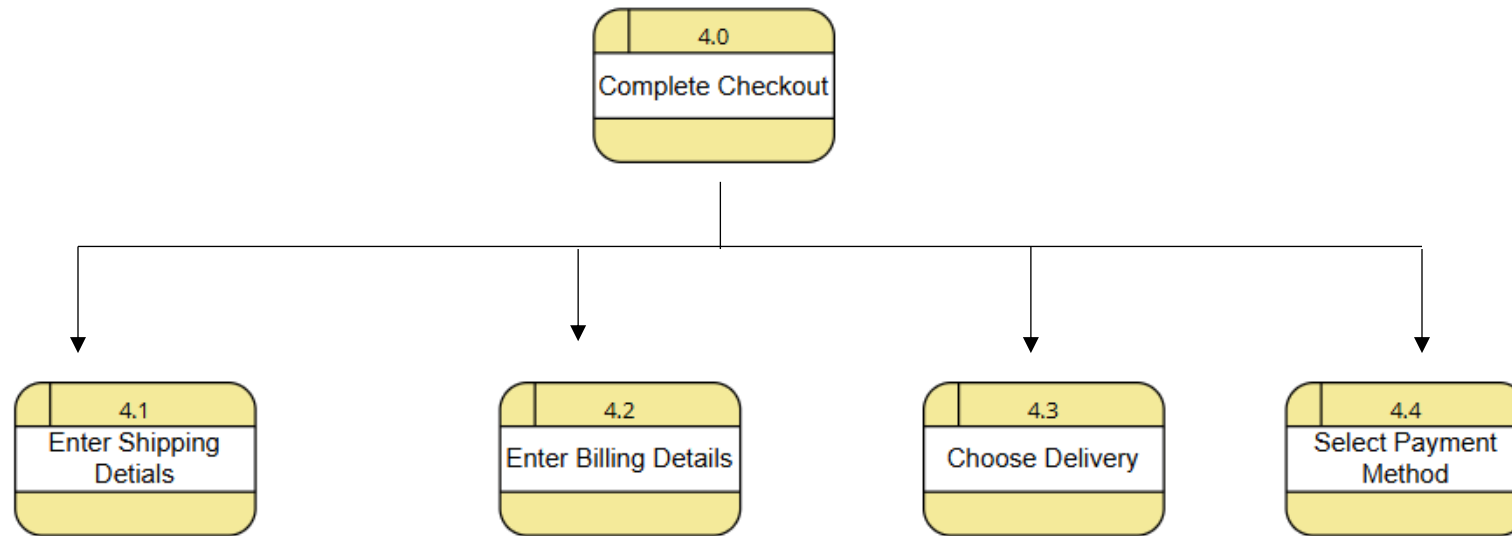


Level 2

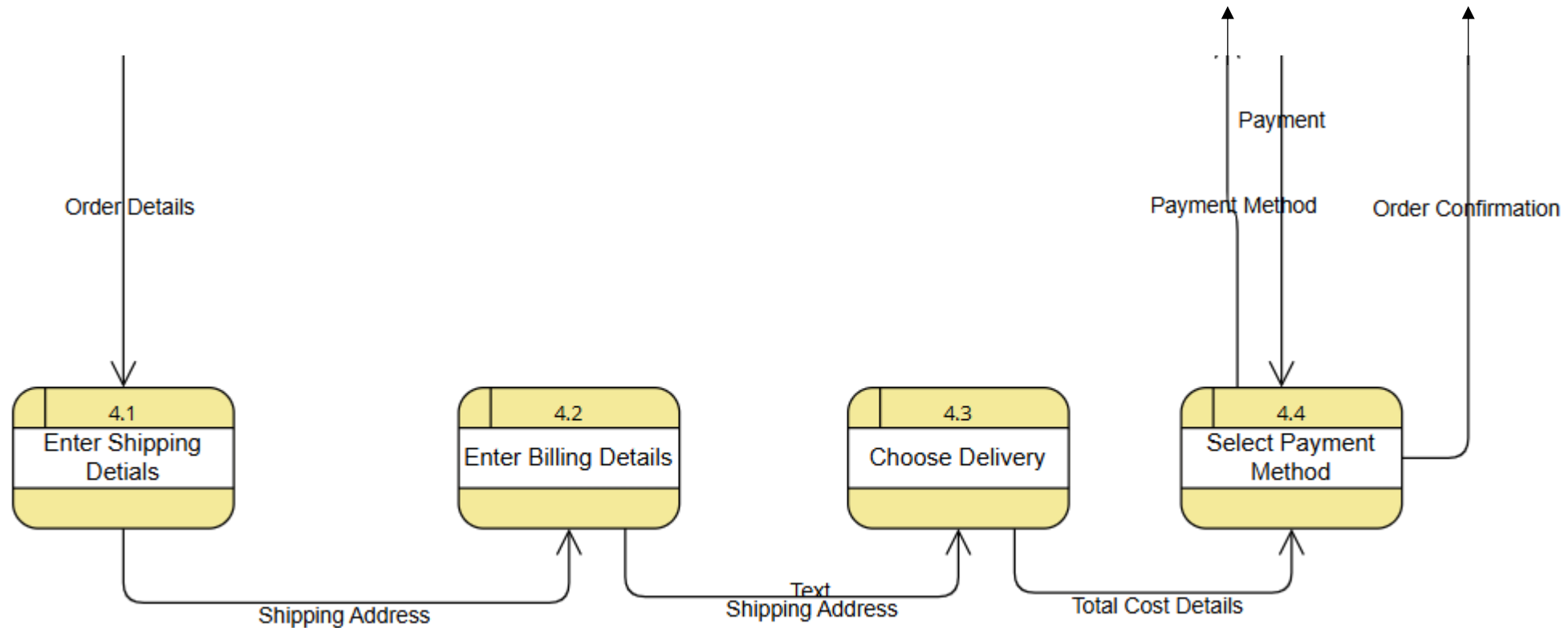
- Level 2 diagram decomposes main processes into sub processes.
- Reference number
 - X.Y <- X – Main process number which is exploded, Y- Sequence of the sub processes



Level 2 – Example (1)

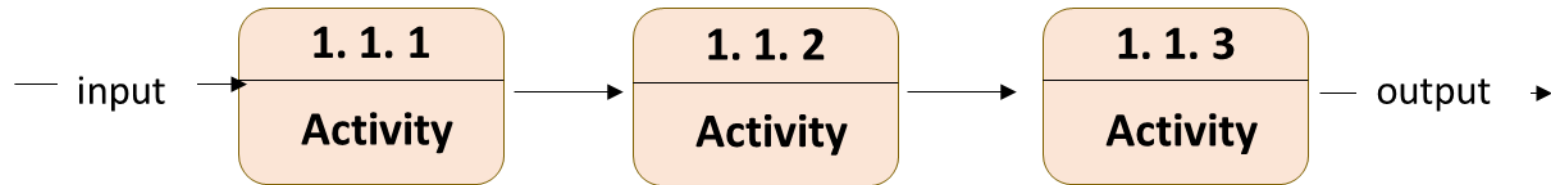
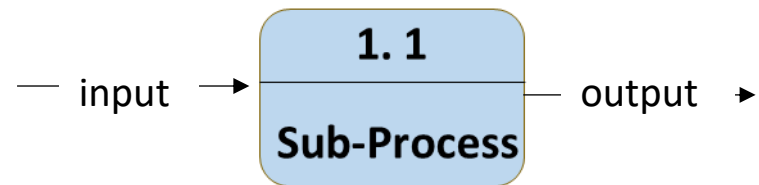


Level 2 – Example (2)



Level 3

- Level 3 decomposes sub processes into more detailed activities.



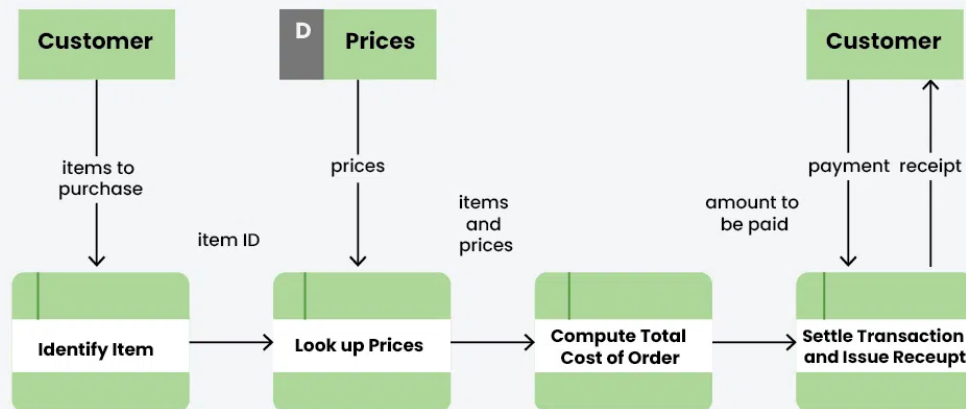
- X.Y.Z<- X – Main process number which is exploded, Y- Sub process, Z- sequence of activities

Data Flow Diagram Types (1)

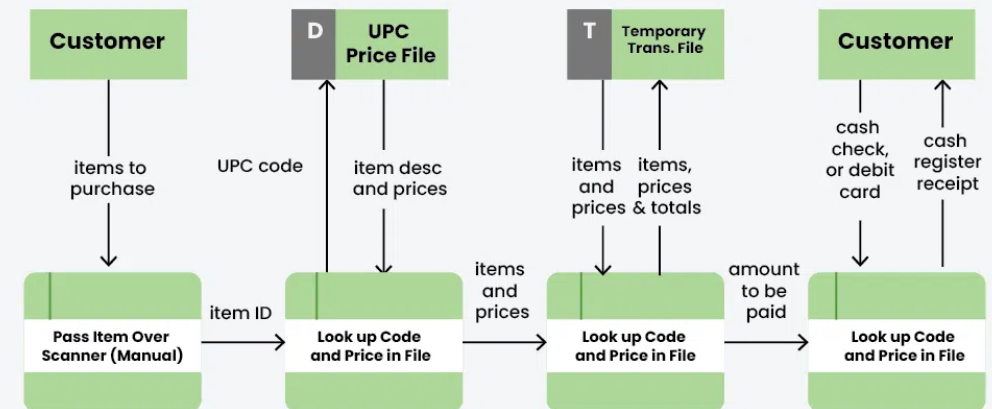
- Logical Data Flow Diagram
 - Logical data flow diagram mainly focuses on the **system process**.
 - It illustrates how data flows in the system.
 - Logical Data Flow Diagram (DFD) mainly focuses on high level processes and data flow without diving deep into technical implementation details.
 - Logical DFD is used in various organizations for the smooth running of system.
 - Business perspective.
 - Business operation goals.
 - Business Analyst or Business System Analyst creates.
- Physical Data Flow Diagram
 - Physical data flow diagram shows how **the data flow is actually implemented in the system**.
 - In the Physical Data Flow Diagram (DFD), we include additional details such as data storage, data transmission, and specific technology or system components.
 - Physical DFD is more specific and close to implementation.
 - System perspective.
 - System implementation.
 - Developer or Database Analyst creates.

Data Flow Diagram Types (2)

Logical Data Flow Diagram (DFD)



Physical Data Flow Diagram (DFD)



Advantages and Disadvantages of Data Flow Diagram (DFD)

- Advantages

- It helps us to understand the functioning and the limits of a system.
- It is a graphical representation which is very easy to understand as it helps visualize contents.
- Data Flow Diagram represent detailed and well explained diagram of system components.
- It is used as the part of system documentation file.
- Data Flow Diagrams can be understood by both technical or nontechnical person because they are very easy to understand.

- Disadvantages

- At times Data Flow Diagram (DFD) can confuse the programmers regarding the system.
- Data Flow Diagram takes long time to be generated, and many times due to this reasons analysts are denied permission to work on it.

Assignment

- A supermarket uses point of sale bar code scanners at their checkout. The bar code on the product is scanned and is then checked to see if it is valid. If it is a valid bar code then the product stock database is updated and an entry for that product is printed as a receipt to the customer. For validation product details are read from product stock database. If it is an invalid bar code then an alarm will sound on the checkout and the operator will need to deal with the problem.
- Draw the context and level 1 Data Flow Diagrams.